

Гиперболические функции

$$\text{Def} \quad \text{sh } x = \frac{e^x - e^{-x}}{2}$$

$$\text{Def} \quad \text{ch } x = \frac{e^x + e^{-x}}{2} \xrightarrow{x \rightarrow 0} 1$$

$$\text{Def} \quad \text{th } x = \frac{\text{sh } x}{\text{ch } x}$$

$$\text{Ymb} \quad \lim_{x \rightarrow 0} \frac{\text{sh } x}{x} = \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} + \frac{e^{-x} - 1}{-x} \right) = 1$$

$$\text{Ymb} \quad \lim_{x \rightarrow 0} \frac{\text{th } x}{x} = 1 \quad \left(\frac{\text{sh } x}{x} \cdot \frac{1}{\text{ch } x} \right)$$

ЭКВИВАЛЕНТ

$$\text{Def} \quad f, g: \dot{U}_\delta(x_0) \rightarrow \mathbb{R}$$

$$\forall x \in \dot{U}_\delta(x_0) \rightarrow f(x) \neq 0, g(x) \neq 0$$

$$f \sim g \text{ при } x \rightarrow x_0, \text{ если } \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1$$

$$\text{Ymb} \quad f, g, h: \dot{U}_\delta(x_0) \rightarrow \mathbb{R} \setminus \{0\}$$

$$\left. \begin{array}{l} f \sim g, \text{ при } x \rightarrow x_0 \\ g \sim h, \text{ при } x \rightarrow x_0 \end{array} \right\} \Rightarrow f \sim h \text{ при } x \rightarrow x_0$$

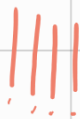
$$\text{Лемма} \quad \frac{f}{h} = \frac{f}{g} \cdot \frac{g}{h}$$

$$\text{Ymb} \quad f_1, f_2, g_1, g_2, h_1, h_2: \dot{U}_\delta(x_0) \rightarrow \mathbb{R} \setminus \{0\}$$

$$\left. \begin{array}{l} f_1 \sim f_2 \text{ при } x \rightarrow x_0 \\ g_1 \sim g_2 \text{ при } x \rightarrow x_0 \\ h_1 \sim h_2 \text{ при } x \rightarrow x_0 \end{array} \right\} \Rightarrow \frac{f_1 g_1}{h_1} \sim \frac{f_2 g_2}{h_2} \text{ при } x \rightarrow x_0$$

Лок-бо

$$\frac{\frac{f_1(x) g_1(x)}{h_1(x)}}{\frac{f_2(x) g_2(x)}{h_2(x)}} = \frac{f_1(x)}{f_2(x)} \cdot \frac{g_1(x)}{g_2(x)} \cdot \frac{h_2(x)}{h_1(x)} \xrightarrow{x \rightarrow x_0} 1$$



$$\sin x \sim \tan x \sim \ln(1+x) \sim e^x - 1 \sim \sinh x \sim \tanh x \sim x \text{ при } x \rightarrow 0$$

Ymb. $f_1 \sim f_2, g_1 \sim g_2 \not\Rightarrow f_1 + g_1 \sim f_2 + g_2$

$$x + x^3 \not\sim x \quad (x \rightarrow 0)$$

$$\downarrow$$

$$-x \sim -x + x^2 \quad (x \rightarrow 0)$$

$$x^3 \sim x^2 \quad (\text{БЕДА})$$

Ymb. $f, g: \dot{U}_\delta(y) \rightarrow \mathbb{R} \setminus \{0\}$
 $f \sim g \text{ при } y \rightarrow y_0$
 $\gamma: \dot{U}_\delta(x_0) \rightarrow \mathbb{R}$
 $\gamma(x) \rightarrow y_0 \text{ при } x \rightarrow x_0$

$$\exists \beta \in (0, \delta]: \forall x \in \dot{U}_\beta(x_0) \rightarrow \gamma(x) \in \dot{U}_\delta(y_0)$$

$$\downarrow$$

$$f(\gamma(x)) \sim g(\gamma(x)) \quad (x \rightarrow x_0)$$

$$\lim_{x \rightarrow x_0} \frac{f(\gamma(x))}{g(\gamma(x))} = \lim_{y \rightarrow y_0} \frac{f(y)}{g(y)} = 1$$

0 ЧАНАЯ

by

$f, g: \dot{U}_\delta(x_0) \rightarrow \mathbb{R} \setminus \{0\}$
 f — бесконечно малая относительно g при $x \rightarrow x_0$
 $f(x) = o(g(x)) \quad (x \rightarrow x_0)$

"0 малая"

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0$$

$$x^2 = o(x) \quad x \rightarrow 0$$

$$x^3 = o(x) \quad x \rightarrow 0$$

$$f, g : \dot{U}_0(v_0) \rightarrow \mathbb{R} \setminus \{0\}$$

$$f \sim g \text{ near } x \rightarrow x_0 \Leftrightarrow f(x) - g(x) = o(g(x)) \text{ near } x \rightarrow x_0$$

$$\text{L'Hôpital: } \frac{f(x)}{g(x)} \xrightarrow{x \rightarrow x_0} \frac{f(x)}{g(x)} - 1 \xrightarrow{x \rightarrow x_0} 0 \quad \frac{f(x) - g(x)}{g(x)} \xrightarrow{x \rightarrow x_0} 0 \Leftrightarrow f(x) - g(x) = o(g(x))$$

$$\sin x = x + o(x)$$

$$t_f = x + o(x) \quad x \rightarrow 0$$

$$e^x = 1 + x + o(x)$$

$$\text{Om } f, g \quad \dot{V}_0(x_0) \rightarrow \mathbb{R}$$

$$f(x) = O(g(x)) \quad x \rightarrow x_0$$

$$\exists C > 0 \exists \delta_0 \in (0, \delta] : \forall x \in \dot{U}_{\delta_0}(x_0) \hookrightarrow |f(x)| \leq C |g(x)|$$

$$\frac{O(O(g(x)))}{O(O(f(x)))} \quad \underbrace{f \cdot O(g)}_h = d(fg) \quad x \rightarrow x_0$$

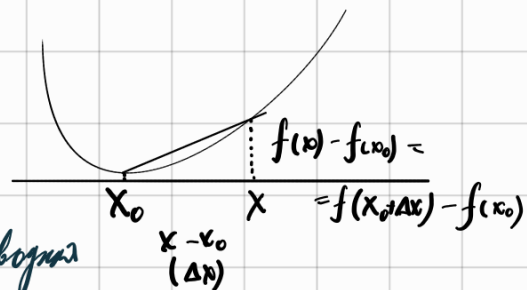
$$\frac{h}{f} = dg \quad \frac{h}{fg} \Rightarrow x \rightarrow x_0$$

$$x^2 \cdot O(x) = O(x^3)$$

$$\lambda^{-1} \cdot O(x) = O(1)$$

$$f(x) = o(1) \quad (x \rightarrow x_0)$$

$$f(x) \rightarrow 0$$


$$\text{Def } f: \mathcal{V}(\mathfrak{v}_0) \rightarrow \mathbb{R}$$

тогда $\exists \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \in \mathbb{R}$, то $\exists$ производная.

функции f в $T. X_0$

Eine Funktion nennt man konvex $f'(x) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$

$$y_{\text{gen}}(x) - f(x) = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} (x - x_0)$$

$$f(x(x, \Delta x)) = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} (x - x_0) + f(x_0)$$

Производная

Опр: касательная к графику $y = f(x)$ в т. $(x_0; f(x_0))$

$$y_{\text{кас}}(x) = k \Delta x (x - x_0) + f(x_0)$$

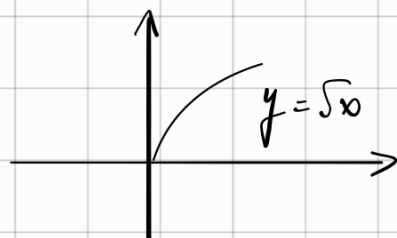
$$f'(x) = \lim_{\Delta x \rightarrow 0} k(\Delta x) (x - x_0) + f(x_0)$$

$$\exists f'(x_0) \in \mathbb{R} \Leftrightarrow \exists \text{ кас. к } y = f(x) \text{ в } (x_0; f(x_0)) \quad y = f'(x_0)(x - x_0) + f(x_0)$$

$$\exists f'(x_0) = \infty \Leftrightarrow \exists \text{ бегм. кас. в } (x_0; f(x_0))$$

$$y - f(x_0) = k \Delta x (x - x_0)$$

$$x - x_0 = \frac{1}{k(\Delta x)} (y - f(x_0))$$



Опр $f: U \subset \mathbb{R} \rightarrow \mathbb{R}$

f дифференцируема в т. x_0 , если $\exists A \in \mathbb{R} : f(x) - f(x_0) = A(x - x_0) + o(x - x_0)$
($x \rightarrow x_0$)

Умб $f: U \subset \mathbb{R} \rightarrow \mathbb{R}$

f дифференцируема в $x_0 \Leftrightarrow \exists f'(x_0) \in \mathbb{R}$
и тогда $A = f'(x_0)$

Док-во:

$$f(x) - f(x_0) = A(x - x_0) + o(x - x_0) \text{ при } x \rightarrow x_0 \Leftrightarrow$$

$$\Leftrightarrow \frac{f(x) - f(x_0)}{x - x_0} = A + o(1) \text{ при } x \rightarrow x_0 \Leftrightarrow$$

$$\Leftrightarrow \frac{f(x) - f(x_0)}{x - x_0} - A = o(1) \text{ при } x \rightarrow x_0 \Leftrightarrow$$

$$\lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} - A \right) = 0 \Leftrightarrow \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = A$$

Опр $\int x = \Delta x$ (дифференциал x), если
 $\exists f'(x_0) \in \mathbb{R} \quad \int f = f'(x_0) \int x$

Ymb $f: \mathcal{U}(x_0) \rightarrow \mathbb{R}$

$$\exists f'(x_0) \in \mathbb{R} \Leftrightarrow \begin{cases} \exists f'_+(x_0) \\ \exists f'_-(x_0) \\ f'_+(x_0) = f'_-(x_0) \end{cases}, \text{vgl}$$

$$f'_+(x_0) = \lim_{x \rightarrow x_0+0} \frac{f(x) - f(x_0)}{x - x_0} \quad f'_-(x_0) = \lim_{x \rightarrow x_0-0} \frac{f(x) - f(x_0)}{x - x_0}$$

Ymb: $f: \mathcal{U}(x_0) \rightarrow \mathbb{R}$

f gupp b.T. $x_0 \Rightarrow f$ reg. b.T. x_0

$$\blacktriangleright f(x) - f(x_0) = f'(x_0)(x - x_0) + o(x - x_0) \text{ mit } x \rightarrow x_0$$

$$\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0 \Rightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0) \text{ reg. reg.}$$

~~Thy~~ $f(x) = |x| \quad f'_+(0) = 1 \quad f'_-(0) = -1$

$$f(x) = \begin{cases} x^2, & x \in \mathbb{R} \setminus \mathbb{Q} \\ 0, & x \in \mathbb{Q} \end{cases} \text{ - reg. b.T. } 0$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \quad \forall \varepsilon > 0 \exists \delta = \varepsilon \quad \forall x \in \mathcal{U}_\delta(0) \hookrightarrow \left| \frac{f(x)}{x} \right| < \delta = \varepsilon$$

Ymb: $f, g: \mathcal{U}(x_0) \rightarrow \mathbb{R} \quad \exists f'(x_0); g'(x_0)$

$$1) (f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$$

$$2) (fg)'(x_0) = f'(x_0) g(x_0) + f(x_0) g'(x_0)$$

$$3) \text{ falls } g(x_0) \neq 0 \quad \left(\frac{f}{g} \right)'(x_0) = \frac{f'(x_0) g(x_0) - f(x_0) g'(x_0)}{g^2(x_0)}$$

$$\blacktriangleright 1) \lim_{x \rightarrow x_0} \frac{(f \pm g)(x) - (f \pm g)(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \pm \frac{g(x) - g(x_0)}{x - x_0} \right)$$

$$= f'(x_0) + g'(x_0)$$

$$2) \lim_{x \rightarrow x_0} \frac{(fg)(x) - fg(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \underbrace{g(x)}_{g(x_0)} \frac{f(x) - f(x_0)}{x - x_0} + f(x_0) \frac{g(x) - g(x_0)}{x - x_0} =$$

$$= f'(x_0)g(x_0) + f(x_0)g'(x_0)$$

$$3) \lim_{x \rightarrow x_0} \frac{\left(\frac{f}{g}\right)(x) - \left(\frac{f}{g}\right)(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{f(x)g(x_0) - f(x_0)g(x)}{g(x)g(x_0)(x - x_0)} =$$

$$= \lim_{x \rightarrow x_0} \frac{1}{g(x)g(x_0)} \left(g(x_0) \frac{f(x) - f(x_0)}{x - x_0} - f(x_0) \frac{g(x) - g(x_0)}{x - x_0} \right) =$$

$$= \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}$$

Ymb. $\left. \begin{array}{l} g: U_{\delta}(x_0) \rightarrow X \\ f: U_{\beta}(y_0) \rightarrow \mathbb{R} \\ g(x_0) = y_0; x \in U_{\delta}(x_0) \\ \exists g'(x_0); f'(y_0) \in \mathbb{R} \end{array} \right\} \Rightarrow$

$$\begin{aligned} h(x) &= f(g(x)) \\ g \text{ sup. b. } x_0 \\ h'(x_0) &= f'(y_0) g'(x_0) \end{aligned}$$

$$\blacktriangle z(y) = \begin{cases} \frac{f(y) - f(y_0)}{y - y_0} & y \neq y_0 \\ f'(y_0), & y = y_0 \end{cases}$$

$$\lim_{y \rightarrow y_0} z(y) = f'(y_0) = z(y_0) \quad z(g(x)) \text{ nemy b. } x_0$$

$$f(y) - f(y_0) = z(y)(y - y_0)$$

$$f(y) - f(g(x_0)) = z(y)(y - g(x_0))$$

$$f(g(x)) - f(g(x_0)) = z(g(x))(g(x) - g(x_0))$$

$$\frac{f(g(x)) - f(g(x_0))}{x - x_0} = z(g(x)) \frac{g(x) - g(x_0)}{x - x_0}$$

$$\lim_{x \rightarrow x_0} \left(z(g(x)) \frac{g(x) - g(x_0)}{x - x_0} \right) = f'(y_0) g'(x_0); \quad h'(x_0) = f'(y_0) g'(x_0)$$

Ymb

- 1) $(a^x)' = \ln a \cdot a^x$
- 2) $(\log_a x)' = \frac{1}{x \ln a}$
- 3) $(x^n)' = n x^{n-1} \quad x \in \mathbb{R} \quad n \in \mathbb{N}$
- 4) $(x^\alpha)' = \alpha x^{\alpha-1} \quad x > 0 \quad \alpha \in \mathbb{R}$
- 5) $(\sin x)' = \cos x$
 $(\cos x)' = -\sin x$
- 6) $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$
 $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$
- 7) $(\operatorname{sh} x)' = \operatorname{ch} x$
 $(\operatorname{ch} x)' = \operatorname{sh} x$

► 1) $\lim_{x \rightarrow x_0} \frac{e^x - e^{x_0}}{x - x_0} = e^{x_0} \lim_{x \rightarrow x_0} \frac{e^{x-x_0} - 1}{x - x_0} = e^{x_0} \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = e^{x_0}$

$$(e^x)' = e^x$$

$$(a^x)' = (e^{\ln a \cdot x})' = e^{(\ln a) \cdot x} (\ln a \cdot x)' = a^x \ln a$$

2) $\lim_{x \rightarrow x_0} \frac{\log_a x - \log_a x_0}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\frac{\ln x}{\ln a} - \frac{\ln x_0}{\ln a}}{x - x_0} = \frac{1}{\ln a} \lim_{x \rightarrow x_0} \frac{\ln \frac{x}{x_0}}{x_0 \left(\frac{x}{x_0} - 1 \right)} =$
 $= \frac{1}{x_0 \ln a} \lim_{y \rightarrow 0} \frac{\ln y + 1}{y} = \frac{1}{x_0 \ln a}$

3) $\lim_{x \rightarrow x_0} \frac{x^n - x_0^n}{x - x_0} = \lim_{x \rightarrow x_0} \frac{(x - x_0)(x^{n-1} + x^{n-2}x_0 + \dots + x x_0^{n-2} + x_0^{n-1})}{x - x_0} = n x_0^{n-1}$

4) $(x^\alpha)' = (e^{(\ln x) \alpha})' = e^{(\ln x) \alpha} ((\ln x) \alpha)' = x^\alpha \cdot \alpha \cdot \frac{1}{x} = \alpha x^{\alpha-1}$

5) $\lim_{x \rightarrow x_0} \frac{\sin x - \sin x_0}{x - x_0} = \lim_{x \rightarrow x_0} \frac{2 \sin \frac{x-x_0}{2} \cos \frac{x+x_0}{2}}{x - x_0} = \cos x_0$

7) $(\operatorname{sh})' x = \left(\frac{e^x - e^{-x}}{2} \right)' = \frac{e^x + e^{-x}}{2} = \operatorname{ch} x$ ▲

Ymb: $f: U_{\delta}(x_0) \rightarrow \mathbb{R}$

f - continuous function

f - invertible at x_0

$f'(x_0) \neq 0$

$$\left. \begin{array}{l} f \text{ - continuous function} \\ f \text{ - invertible at } x_0 \\ f'(x_0) \neq 0 \end{array} \right\} \Rightarrow \begin{array}{l} g(y) = f'(y) \\ y = f(x_0), \text{ mo } g'(y_0) = \frac{1}{f'(x_0)} \end{array}$$

$$\lim_{y \rightarrow y_0} \frac{g(y) - g(y_0)}{y - y_0} = \lim_{x \rightarrow x_0} \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} = \lim_{x \rightarrow x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{f'(x_0)}$$

$x = g(y)$
 $y = f(x)$
 $g'(y_0) = \frac{1}{f'(x_0)}$

6) $y_0 = \arcsin x_0$; $x_0 = \sin y_0$

$$(\arcsin)'(x_0) = \frac{1}{\sin' y_0} = \frac{1}{\cos(\arcsin x_0)} = \frac{1}{\sqrt{1 - \sin^2 \arcsin x_0}} = \frac{1}{\sqrt{1 - x_0^2}}$$

$$(tg x)' = \frac{1}{\cos^2 x}$$

$$(\arctg)'(x_0) = \frac{1}{tg'(y_0)} = \cos^2 y_0 = \frac{1}{1 + tg^2 y_0} = \frac{1}{1 + x_0^2}$$

$$F(x, y) = 0$$

$$x^2 - y^2 = 1 \Rightarrow y = \sqrt{x^2 - 1}$$

$$y = f(x)$$

$$(F(x, f(x)))'$$

$$y = -\sqrt{x^2 - 1}$$

$$x^2 - y^2 - 1 = 0$$

$$2x - 2yy' = 0 \text{ — линейно относительно } y'$$

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

$$y = y(t(x))$$

КУДА, ЧТОБЫ

$x(t)$ БЫЛА

ОБРАТИЛА

$$\Rightarrow y'_x = (y(t(x)))' = y'_t \cdot t'_x = \frac{y'_t}{x'_t}$$

$$y''_{xx} = \frac{(y'_t)_t}{x'_t} = \frac{\left(\frac{y'_t}{x'_t}\right)'_t}{x'_t} = \frac{y''_{tt} x'_t - y'_t x''_{tt}}{(x'_t)^3}$$

ОСТАНАВЛИВАТЬСЯ
ЛУЧШЕ ЗДЕСЬ



Опр. $f^{(0)}(x) = f(x) \leftarrow y \text{ не ОПРЕДЕЛЕНА}$

$\exists$ в $U_\delta(x_0)$ задана $f^{(n)}(x)$, ТОГДА $f^{(n+1)}(x_0) = (f^{(n)})'(x_0)$

Умб. $\exists f^{(n)}(x_0) \Rightarrow \exists \delta: U_\delta(x_0)$ n -м $f^{(k)}(x)$ определены при $k < n$

Оп. $\int f = f'(x_0) dx$
 $\int^{n+1} f(x_0) = \int (\int f''(x_0))$

Умб $\exists \int^n f(x_0) \Leftrightarrow \exists f^{(n)}(x_0) \in \mathbb{R}$. Более того, если $\exists f''(x_0)$, то
 $\int^n(f(x_0)) = f^{(n)}(x_0) (dx)^n$

► при $n=1$ по оп.

$\exists k \in \mathbb{N} \exists U_\delta(x_0): f^{(k)}(x_0)$ - определена $\Rightarrow \exists f^{(k+1)}(x_0), \exists \int^k f(x)$ и в $\int^1 \int^{k+1} f(x_0)$ окр. ???

$\exists \exists U_\delta(x_0) f^{(k)}(x)$ - оп. $\Rightarrow \int^{(k)} f(x)$ - оп. и $\int^{(k)} f(x) = f^{(k)}(x) dx^k$
 $\int (\int^k f(x)) \Big|_{x=x_0} = \int (f^{(k)}(x) dx^k) \Big|_{x=x_0} = dx^k \cdot \int (f^{(k)}(x)) =$
 $= f^{(k+1)}(x_0) dx^{k+1} \Rightarrow f^{(k+1)}(x_0) dx^{k+1}$ A

Оп. Будем говорить, что f -групп и раз в т. x_0 , если $\exists \int^n f(x_0)$

$\int(f(y)) = f'(y) dy$
 $\int f(y(x)) = f(y) y'(x) dx = f'(y) dy(x) \Rightarrow$ ф-ла первого групп инварианта в зависимости y

$\int^2 f(y) = f''(y) dy^2$

$\int^2 f(y(x)) = \int (f''(y) y'(x) dx) = dx (\int (f''(y)) y'(x) + f'(y) \cdot \int (y'(x)) =$
 $= (f''(y)(y'(x))^2 + f'(y) y''(x)) dx^2$

$\int^2 f(y) = f''(y) dy^2$ - не совпадают

$\int (f'(y) dy) = \int (f'(y)) dy + \int f'(y) dy = f''(y) dy^2 + f'(y) dy$

Оп. $f: X \rightarrow \mathbb{R}$

$x_0 \in X$ т. лок min, если $\exists \delta > 0: \forall x \in X \cap U_\delta(x_0) \rightarrow f(x) \geq f(x_0)$
 лок max, если $\exists \delta > 0: \forall x \in X \cap U_\delta(x_0) \rightarrow f(x) \leq f(x_0)$

x_0 - т. нестрого лок экстр., если x_0 - т. лок min или max

x_0 - т. строго лок min, если $\exists \delta > 0 \forall x \in U_\delta(x_0) \rightarrow f(x) > f(x_0)$
 max $f(x) < f(x_0)$

x_0 - т. стр. лок экстр., если x_0 - т. строг лок max/min

Умб (Т. Ферма)

$$\left. \begin{array}{l} f: (a; b) \rightarrow \mathbb{R} \\ x_0 - \text{Т. ЛОК ЭКСТР} \\ f - \text{гипс в Т. } x_0 \end{array} \right\} \Rightarrow f'(x_0) = 0$$

БОО $x_0 - \text{Т. ЛОК. max}$

$$\left. \begin{array}{l} f'_+(x_0) = \lim_{x \rightarrow x_0+0} \frac{f(x) - f(x_0)}{x - x_0} \leq 0 \\ f'_-(x_0) = \lim_{x \rightarrow x_0-0} \frac{f(x) - f(x_0)}{x - x_0} \geq 0 \end{array} \right\} \Rightarrow f'(x_0) = 0$$

Умб. (Т. Ролл)

$$f: [a; b] \rightarrow \mathbb{R}$$

$$\left. \begin{array}{l} 1) f \text{ непрерывна на } [a; b] \\ 2) f \text{ гипс на } (a; b) \\ 3) f(a) = f(b) \end{array} \right\} \Rightarrow \exists \xi \in (a; b) : f'(\xi) = 0$$

по Т. Вейерштрасса $\exists \min_{[a; b]} f(x) \max_{[a; b]} f(x)$

$$\left. \begin{array}{l} 1) \text{ Если Т. min и Т. max - концы отрезка } \Rightarrow f(x) = c \Rightarrow \forall x \in (a; b) \hookrightarrow f'(x) = 0 \\ 2) \text{ Т. min или Т. max } \in (a; b) \\ \text{БОО } x_0 - \text{Т. min; } x_0 \in (a; b) \end{array} \right\} \xrightarrow{\text{по Т. Ферма}} f'(x_0) = 0$$

Умб (Т. Коши)

$$\left. \begin{array}{l} f, g: [a; b] \rightarrow \mathbb{R} \\ f, g \text{ непрерывны на } [a; b] \\ f, g \text{ гипс на } (a; b) \\ \forall x \in (a; b) \hookrightarrow g'(x) \neq 0 \end{array} \right\} \Rightarrow \exists \xi \in (a; b) : \frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Если $g(a) = g(b) \Rightarrow$ по Т. Ролла $\exists x_0 \in (a; b) : g'(x_0) = 0$ противо речие $\Rightarrow g(a) \neq g(b)$

на $[a; b]$ - отрезок $\psi(x) = f(x) - \lambda g(x)$, где $\lambda \in \mathbb{R}$

ψ - непрерывна на $[a; b]$ и гипс на $(a; b)$

хотим, чтобы $\psi(a) = \psi(b) \Rightarrow f(a) - \lambda g(a) = f(b) - \lambda g(b) \Rightarrow \lambda = \frac{f(b) - f(a)}{g(b) - g(a)} = \lambda_0 \Rightarrow$

по Т. Ролля

$$\Rightarrow \exists \xi \in (a; b) : \psi'(\xi) = 0$$

$$\psi(x) = f(x) - \lambda_0 g(x)$$

$$\psi'(x) = f'(x) - \lambda_0 g'(x) \quad f'(\xi) - \lambda_0 g'(\xi) = 0$$

$$\frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Умб (Т. ЛАГРАНЖА)

$$\left. \begin{array}{l} f: [a; b] \rightarrow \mathbb{R} \\ f \text{ непрерывна на } [a; b] \\ f \text{ дифференцируема на } (a; b) \end{array} \right\} \Rightarrow \exists \xi \in (a; b) : f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow \text{по Т. Коши} \quad \exists \xi \in (a; b) : \frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Ом $f: U_\delta(x_0) \rightarrow \mathbb{R}$

$$\exists f^{(k)}(x_0) \in \mathbb{R}$$

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad \text{— многочлен Тейлора}$$

$r_n(x) = f(x) - P_n(x)$ — остаточный член n -го порядка в ф. Тейлора

$\Downarrow$

$$f(x) = P_n(x) + r_n(x)$$

$$\text{Умб. } \psi_n(x) = (x - x_0)^n \Rightarrow 1) \psi_n^{(k)}(x) = \begin{cases} 0 & k \geq n+1 \\ \frac{n!}{(n-k)!} (x - x_0)^{n-k}, & k \in \{0, 1, \dots, n\} \end{cases}$$

$$2) \psi_n^{(k)}(x_0) = \begin{cases} n! & k = n \\ 0 & k \neq n \end{cases}$$

$$\Rightarrow \text{при } k=0 \quad \psi_n^{(0)}(x) = \frac{n!}{(n-0)!} (x - x_0)^{n-0}$$

$$\} \quad k < n \quad \psi_n^{(k)}(x) = \frac{n!}{(n-k)!} (x - x_0)^{n-k}$$

$$\begin{aligned} \psi_n^{(k+1)}(x) &= (\psi_n^{(k)}(x))' = \frac{n!}{(n-k)!} ((x - x_0)^{n-k})' = \frac{n!}{(n-k)!} (n-k) (x - x_0)^{n-(k+1)} \\ &= \frac{n!}{(n-(k+1))!} (x - x_0)^{n-(k+1)} \end{aligned}$$

$$\psi_n^{(n-1)} = n! (x - x_0) \quad \psi_n^{(n)}(x) = n!$$

$$\forall m \exists f^{(m)}(x_0) \in \mathbb{R} \Rightarrow \forall k \in \{0; \dots; n\} \rightarrow r_n^{(k)}(x_0) = 0$$

$$r_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$$

$$\triangleright r_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} \psi_k(x)$$

$$r_n^{(m)}(x) = f^{(m)}(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} \psi_k^{(m)}(x)$$

$$m \in \{0; \dots; n\}$$

$$r_n^{(n)}(x) = f^{(n)}(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} \psi_k^{(n)}(x)$$

$$r_n^{(n)}(x_0) = f^{(n)}(x_0) - \frac{f^{(n)}(x_0)}{n!} = 0 \Rightarrow r_n^{(m)}(x_0) = 0$$

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + r_n(x)$$

$$(\sin x)''' = (\cos x)'' = (-\sin x)' = -\cos x$$

$$\sin x = x - \frac{1}{6} x^3 + r_n(x) \quad \leftarrow \text{НЕ ЗНАЕМ ЧТО ЗА } \varphi\text{-УЗ}$$

$\forall m \exists$ (Ф-ЛА ТЕЙЛОРА С ОСТАТОЧНЫМ ЧЛЕНОМ В ФОРМЕ ПЕАНО)

$$\left. \begin{array}{l} f: U_{\delta_0}(x_0) \rightarrow \mathbb{R} \\ \exists f^{(n)}(x_0) \in \mathbb{R} \end{array} \right\} \Rightarrow f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + \underbrace{o((x-x_0)^n)}_{r_n(x)} \quad x \rightarrow x_0$$

$$\sin x = x + o(x^2)$$

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

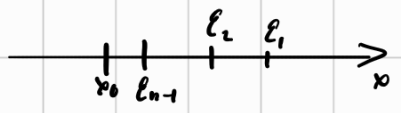
$\triangleright \exists \delta \in (0; \delta_0] : \forall U_{\delta}(x_0) \text{ сущ. } f^{(k)}(x) \text{ при } k \leq n-1$

$$\frac{r_n(x)}{(x-x_0)^n} = \frac{r_n(x) - r_n(x_0)}{\psi_n(x) - \psi_n(x_0)} \stackrel{\text{по т. КОШИ}}{=} \exists \xi_1 \text{ из ИНТЕРВАЛА } x \text{ и } x_0 : \frac{r_n'(\xi_1) - r_n'(x_0)}{\psi_n'(\xi_1) - \psi_n'(x_0)} =$$

(хз кто ξ_1)

$$\stackrel{\text{по т. КОШИ}}{=} \frac{r_n''(\xi_2)}{\psi_n''(\xi_2)} = \dots = \frac{r_n^{(n-1)}(\xi_{n-1}) - r_n^{(n-1)}(x_0)}{\psi_n^{(n-1)}(\xi_{n-1}) - \psi_n^{(n-1)}(x_0)} = \frac{1}{n!} \frac{r_n^{(n)}(\xi_{n-1}) - r_n^{(n)}(x_0)}{\xi_{n-1} - x_0}$$

$$\lim_{x \rightarrow x_0} \frac{r_n(x)}{(x-x_0)^n} = \lim_{x \rightarrow x_0} \frac{r_n^{(n-1)}(\xi_1) - r_n^{(n-1)}(x_0)}{n! (\xi_{n-1} - x_0)} = \lim_{\xi_{n-1} \rightarrow x_0} \frac{r_n^{(n-1)}(\xi_{n-1}) - r_n^{(n-1)}(x_0)}{n! (\xi_{n-1} - x_0)} =$$

$$= \frac{1}{n!} r_n^{(n)}(x_0) = 0 \Rightarrow r_n(x) = o((x-x_0)^n)$$


Умб. (Формула Тейлора составленная членом в форме Лагранжа)

$$f: U_{\delta_0}(x_0) \rightarrow \mathbb{R}$$

$\exists f^{(n+1)} \in U_{\delta}(x_0) \Rightarrow \forall x \in U_{\delta}(x_0) \exists \xi$ между x и x_0 :

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}}_{r_n(x)}$$

$$\begin{aligned} \frac{r_n(x)}{(x-x_0)^{n+1}} &= \frac{r_n(x) - r_n(x_0)}{\psi_{n+1}(x) - \psi_{n+1}(x_0)} \stackrel{\text{по т. Кольца}}{=} \frac{r_n'(\xi_1) - r_n'(x_0)}{\psi_{n+1}'(\xi_1) - \psi_{n+1}'(x_0)} \stackrel{\text{по т. Кольца}}{=} \frac{r_n''(\xi_2)}{\psi_{n+1}''(\xi_2)} = \dots = \frac{r_n^{(n+1)}(\xi)}{\psi_{n+1}^{(n+1)}(\xi)} = \\ &= \frac{f^{(n+1)}(\xi)}{(n+1)!} \end{aligned}$$

Умб. Т. единственности разложения Ф-и по формуле Тейлора

$$f: U_{\delta_0}(x_0) \rightarrow \mathbb{R}$$

$$\exists f^{(n)}(x_0) \in \mathbb{R}$$

$$f(x) = \sum_{k=0}^n a_k (x-x_0)^k + o((x-x_0)^n) \text{ при } x \rightarrow x_0$$

$$\Rightarrow \forall k \in \{0, \dots, n\} \hookrightarrow a_k = \frac{f^{(k)}(x_0)}{k!}$$

$$\begin{aligned} f(x) &= \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + o((x-x_0)^n) \text{ при } x \rightarrow x_0 \\ \cancel{f(x_0)} + \frac{\cancel{f'(x_0)}}{1!} (x-x_0) + \frac{f''(x_0)}{2!} (x-x_0)^2 + \dots + o((x-x_0)^n) &= \\ = \cancel{f(x_0)} + a_1 (x-x_0) + a_2 (x-x_0)^2 + \dots + o((x-x_0)^n) \end{aligned}$$

$$f(x_0) = a_0$$

$$\cancel{\frac{f'(x_0)}{1!}} + \frac{f'(x_0)}{2!} (x-x_0) + \dots + o((x-x_0)^{n-1}) =$$

$$= \cancel{a_1} + a_2 (x-x_0) + \dots + o((x-x_0)^{n-1})$$

$$\frac{f'(x_0)}{1} = a_1$$

$$\text{Умб. } f: U_{\delta_0}(x_0) \rightarrow \mathbb{R}$$

$$\exists f^{(n)}(x_0) \in \mathbb{R}$$

$$f(x) = \sum_{k=0}^n a_k (x-x_0)^k + o((x-x_0)^n) \text{ при } x \rightarrow x_0$$

$$\Rightarrow f'(x) = \sum_{k=0}^n k a_k (x-x_0)^{k-1} + o((x-x_0)^{n-1}) \text{ при } x \rightarrow x_0$$

$$\begin{aligned}
 & \triangleright g(x) = f'(x) \\
 & \exists g^{(n-1)}(x_0) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + o((x-x_0)^{n-1}) \quad a_{k+1} = \frac{f^{(k+1)}(x_0)}{(k+1)!} \\
 & b_k = \frac{f^{(k+1)}(x_0)}{k!} = \frac{(k+1) \frac{f^{(k+1)}(x_0)}{(k+1)!}}{(k+1)!} = (k+1) a_{k+1} \\
 & b_{k-1} = k \cdot a_k
 \end{aligned}$$

$$\begin{aligned}
 \text{ymb: } & f: U_{x_0}^{\delta}(x_0) \rightarrow \mathbb{R} \\
 & \exists f^{(n)}(x_0) \in \mathbb{R} \\
 & f' = \sum_{k=0}^{n-1} a_k (x-x_0)^k + o((x-x_0)^{n-1}) \quad \Rightarrow f(x) = \sum_{k=0}^{n-1} \frac{a_k}{k+1} (x-x_0)^{k+1} + o((x-x_0)^n) + f(x_0)
 \end{aligned}$$

$$\begin{aligned}
 \text{ymb } C_{n+1}^k &= C_n^k + C_n^{k-1} \\
 C_n^k &= \frac{n!}{k!(n-k)!}
 \end{aligned}$$

$$\triangleright C_n^k + C_n^{k-1} = \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} = \frac{n!}{k!(n-k+1)!} ((n-k+1) + k) = \frac{(n+1)!}{k!(n+1-k)!} = C_{n+1}^k$$

ymb (Ф-ла Лейбница)

$$\begin{aligned}
 & f, g: X \rightarrow \mathbb{R} \\
 & f, g \text{ и все их дерр на } X \quad \Rightarrow (f \cdot g)^{(n)} = \sum_{k=0}^n C_n^k f^{(k)}(x) \cdot g^{(n-k)}(x)
 \end{aligned}$$

$$\triangleright \text{при } n=1 \quad (f \cdot g)' = f'g + fg'$$

$$\exists (f \cdot g)^{(n)} = \sum_{k=0}^n C_n^k f^{(k)} g^{(n-k)}$$

$$\begin{aligned}
 (f \cdot g)^{(n+1)} &= ((f \cdot g)^{(n)})' = \left(\sum_{k=0}^n C_n^k f^{(k)} g^{(n-k)} \right)' = \sum_{k=0}^n C_n^k (f^{(k)} g^{(n-k)})' = \\
 &= \sum_{k=0}^n C_n^k f^{(k+1)} g^{(n-k)} + \sum_{k=0}^n C_n^k f^{(k)} g^{(n-k+1)} = \\
 &= \sum_{m=1}^{n+1} C_n^{m-1} f^{(m)} g^{(n-m+1)} + \sum_{m=0}^n C_n^m f^{(m)} g^{(n-m+1)} = \sum_{m=0}^{n+1} C_{n+1}^m f^{(m)} g^{(n-m+1)}
 \end{aligned}$$

